Category Theory for Mathematical Physics — Exercise Sheet 10

Modular tensor categories and anyon models

Variant: Hints and answers

Conventions as in Lecture 10: $\tilde{S}_{ab} = \sum_c N_{a\bar{b}}^c \, d_c \, \theta_c/(\theta_a\theta_b)$ (Proposition 10.3), $S = \tilde{S}/\sqrt{\dim C}$, $T = \mathrm{diag}(\theta_a)$, with the canonical $d_a > 0$; $\lambda_a(b) = \tilde{S}_{ab}/d_a$ is the loop of b around a (Equation (10.2)); ω_0 is the killing loop of Lemma 10.6 (not a 3-cocycle). Here c is the central charge modulo 8 (Theorem 10.9), never the braiding, and C is charge conjugation. $\varphi = (1+\sqrt{5})/2$. For $\mathrm{SU}(2)_k$ the labels are spins $j = 0, \frac{1}{2}, \ldots, \frac{k}{2}$ with $d_j = [2j+1]_q$ and $\theta_j = e^{2\pi i\, j(j+1)/(k+2)}$ (Theorem 9.9, after the flip). The toric code's anyons are $1, e, m, \varepsilon$, as in Sheet 7.

Exercise 1 ($(ST)^3$ for Fibonacci). Take Fibonacci's S (Example 10.11), $\theta = \theta_\tau = e^{4\pi i/5}$ and $T = \mathrm{diag}(1, \theta)$.

(i) Compute $\mathrm{tr}(ST)$ and $\det(ST)$, and show that $(\mathrm{tr}\, ST)^2 = \det(ST)$.

(ii) Deduce from Cayley–Hamilton that $(ST)^3 = \dfrac{\theta(1-\theta)}{\sqrt{2+\varphi}}\, 1$, and show that this number is $e^{7\pi i/10}$.

(iii) Compare with $(ST)^3 = e^{2\pi i c/8} S^2$ and the central charge of Example 10.13.

Hint. (i) $\det S = -1$; $2\cos(4\pi/5) = -\varphi$. (ii) For a 2×2 matrix, $M^2 = (\mathrm{tr}M)M - (\det M)1$; and $1 - e^{i\alpha} = -2i\sin(\alpha/2)e^{i\alpha/2}$.

Answer. (i) $\mathrm{tr}\, ST = (1-\theta)/\sqrt{2+\varphi}$ and $\det(ST) = -\theta$; $(\mathrm{tr}\, ST)^2 = -\theta$ because $\theta + \theta^{-1} = -\varphi$. (ii) $(ST)^3 = \theta(1-\theta)(2+\varphi)^{-1/2}\, 1 = e^{7\pi i/10}\, 1$. (iii) $S^2 = 1$ and $2\pi c/8 = 7\pi/10$ for $c = \frac{14}{5}$.

Exercise 2 (The toric code). Take $1, e, m, \varepsilon$ with $\mathbb{Z}_2 \times \mathbb{Z}_2$ fusion ($e \otimes m = \varepsilon$, each its own dual), $d = 1$ and $\theta = (1, 1, 1, -1)$.

(i) Compute S from Proposition 10.3, check $S^2 = 1$, and conclude that the toric code is modular.

(ii) Recover $e \otimes m \cong \varepsilon$ from Verlinde's formula (Theorem 10.8).

(iii) Compute the Gauss sum and c; and the loop $\omega_0 = \frac{1}{4}(1 + e + m + \varepsilon)$ around each anyon.

Hint. (i) In an abelian model with self-dual anyons $\tilde{S}_{ab} = \theta_{a\otimes b}/(\theta_a\theta_b)$. (iii) The loop of b around a is $\tilde{S}_{ab}$ here.

Answer. (i) $S = \frac{1}{2}\left(\begin{smallmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{smallmatrix}\right)$, $S^2 = 1$: invertible. (ii) The rows $2S_{e\cdot}$ and $2S_{m\cdot}$ multiply entrywise to $2S_{\varepsilon\cdot}$, so $N_{em}^c = \delta_{c,\varepsilon}$ by orthogonality. (iii) $1 + 1 + 1 - 1 = 2 = \sqrt{4}\, e^0$: $c = 0$; ω_0 gives 1 around 1 and 0 around e, m, ε.

Exercise 3 ($\mathrm{SU}(2)_2$ and $\mathrm{SU}(2)_3$). (i) For $k = 2$, compute S from Proposition 10.3 with $d = (1, \sqrt{2}, 1)$ and $\theta = (1, e^{3\pi i/8}, -1)$, and compare with the table $S_{jj'} = \sqrt{2/(k+2)}\, \sin\big((2j+1)(2j'+1)\pi/(k+2)\big)$.

(ii) Use Verlinde's formula at $k = 2$ to recover $\frac{1}{2} \otimes \frac{1}{2} \cong 0 \oplus 1$, and the Gauss sum to find c.

(iii) For $k = 3$, write S from the table and use Verlinde's formula to recover $1 \otimes 1 \cong 0 \oplus 1$.

Hint. (i) $\frac{1}{2} \otimes \frac{1}{2}$ has channels $0, 1$; $\frac{1}{2} \otimes 1 \cong \frac{1}{2}$; $1 \otimes 1 \cong 0$. (iii) $\sin(4\pi/5) = \sin(\pi/5)$, $\sin(3\pi/5) = \sin(2\pi/5)$, and $\sin(2\pi/5)/\sin(\pi/5) = \varphi$.

Answer. (i) $S = \frac{1}{2}\left(\begin{smallmatrix} 1 & \sqrt{2} & 1 \\ \sqrt{2} & 0 & -\sqrt{2} \\ 1 & -\sqrt{2} & 1 \end{smallmatrix}\right)$, the table at $k = 2$. (ii) $N_{\frac{1}{2}\frac{1}{2}}^{c} = 1, 0, 1$ for $c = 0, \frac{1}{2}, 1$; $1 + 2e^{3\pi i/8} - 1 = 2e^{3\pi i/8}$: $c = \frac{3}{2}$. (iii) With $a = \sin(\pi/5)$, $b = \sin(2\pi/5)$, $S = \sqrt{\frac{2}{5}}\left(\begin{smallmatrix} a & b & b & a \\ b & a & -a & -b \\ b & -a & -a & b \\ a & -b & b & -a \end{smallmatrix}\right)$; $N_{11}^{c} = 1, 0, 1, 0$.

Exercise 4 (The central charges $\nu/2$). Take the Ising-type data: $1, \sigma, \psi$ with $d = (1, \sqrt{2}, 1)$, $\theta_\sigma = e^{i\pi\nu/8}$ for odd ν, $\theta_\psi = -1$ (Theorem 7.9, Remark 10.15).

(i) Show that $c = \nu/2$ modulo 8.

(ii) Show that S is the same matrix for every odd ν, and invertible. What, then, tells the eight categories apart?

(iii) Find c for the semion ($d = (1, 1)$, $\theta = (1, i)$), and check it against $3k/(k+2)$ at $k = 1$.

Hint. (i) The Gauss sum has three terms. (ii) $\tilde{S}_{\sigma\sigma} = (\theta_1 + \theta_\psi)/\theta_\sigma^2$.

Answer. (i) $1 + 2e^{i\pi\nu/8} - 1 = 2e^{i\pi\nu/8} = \sqrt{4}\, e^{2\pi i(\nu/2)/8}$. (ii) $S = \frac{1}{2}\left(\begin{smallmatrix} 1 & \sqrt{2} & 1 \\ \sqrt{2} & 0 & -\sqrt{2} \\ 1 & -\sqrt{2} & 1 \end{smallmatrix}\right)$, $S^2 = 1$; T (equivalently c) does. (iii) $1 + i = \sqrt{2}\, e^{i\pi/4}$: $c = 1 = 3 \cdot 1/3$.

Exercise 5 (Loops and the killing loop). (i) Draw the loop of b around a strand a (Equation (10.2)). For Fibonacci, compute $\lambda_\tau(\tau)$, and check the character relation $\lambda_\tau(\tau)^2 = 1 + \lambda_\tau(\tau)$ of Lemma 10.4.

(ii) Compute the killing loop $\omega_0 = (1 + \varphi\,\tau)/(2 + \varphi)$ around 1 and around τ.

(iii) In sVec ($1, \psi$; $d = (1, 1)$; $\theta_\psi = -1$; $\psi \otimes \psi \cong 1$) compute $\lambda_\psi(\psi)$ and the killing loop around ψ. Which step of the proof of Lemma 10.6 fails, and what does it say about S?

Hint. (i) $\tilde{S}_{\tau\tau} = -1$ (Example 10.11). (iii) A transparent object has $\lambda_x(b) = d_b$.

Answer. (i) $\lambda_\tau(\tau) = -1/\varphi$, and $\varphi^{-2} = 1 - \varphi^{-1}$. (ii) 1 around 1, $(1 - 1)/(2 + \varphi) = 0$ around τ. (iii) $\lambda_\psi(\psi) = 1$ and the killing loop is 1 around ψ: the step "transparent, hence $x = 1$" fails; $S = \frac{1}{\sqrt{2}}\left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right)$ is singular.

Exercise 6 ⋆. *The* $\mathrm{SU}(2)_k$ *S-matrix from loops.* Lecture 10 cites $S_{jj'} = \sqrt{2/(k+2)}\,\sin\big((2j+1)(2j'+1)\pi/(k+2)\big)$. Derive it. Write $\vartheta = \pi/(k+2)$ and $y = (2j'+1)\vartheta$.

(i) From Proposition 10.3, show that the loop of spin $\frac{1}{2}$ around j' is $\lambda_{j'}(\frac{1}{2}) = 2\cos y$.

(ii) Using that $\lambda_{j'}$ is a character (Lemma 10.4) and that $[\frac{1}{2}]$ generates the fusion ring (Sheet 5, Exercise 2), show that $\lambda_{j'}(j) = \sin\big((2j+1)y\big)/\sin y$.

(iii) Conclude $\tilde{S}_{jj'} = \sin\big((2j+1)(2j'+1)\vartheta\big)/\sin\vartheta$, and with Sheet 5's global dimension the formula for S. Why must the pivotal flip of Theorem 9.9 be made first?

Hint. (i) The channels of $\frac{1}{2} \otimes j'$ are $j' \pm \frac{1}{2}$; compute $\theta_c/(\theta_{1/2}\theta_{j'})$ for each. (ii) The recursion $\lambda(j + \frac{1}{2}) = \lambda(\frac{1}{2})\lambda(j) - \lambda(j - \frac{1}{2})$.

Answer. (i) $\tilde{S}_{\frac{1}{2}j'} = \big(\sin(y+\vartheta)e^{i(y-\vartheta)} + \sin(y-\vartheta)e^{-i(y+\vartheta)}\big)/\sin\vartheta = \sin 2y/\sin\vartheta = 2\cos y\, d_{j'}$. (ii) $\sin((2j+1)y)/\sin y$ satisfies the same recursion with the same start. (iii) $\tilde{S}_{jj'} = d_{j'}\lambda_{j'}(j)$; $\dim = (k+2)/(2\sin^2\vartheta)$; the formula needs every $d > 0$, and the TL $d_{1/2} < 0$ (substituted anyway, it gives signs $(-1)^{2j+2j'}$).